

Thien Nguyen

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SUMMARY

Engineer with 4.5+ years of professional experience in systems integration, pursuing a mechanical engineering role.

Favorite References: Product Design for Manufacture and Assembly (Geoffrey B., et. al.)

CERTIFICATIONS

Certified SolidWorks Professional (**CSWP**), Geometric Dimensioning & Tolerancing Fundamentals (**GD&T Fundamentals**), Project Management Professional (**PMP**)

TECHNICAL SKILLS

Projects: CAD Modeling, Finite Element Analysis (FEA), Design for Manufacturing & Assembly (DFMA), 3D Printing, Prototypes & Manufacturing, Mechanical Testing & Validation, Mechatronics Systems, Programming (Python, C, C++), Product Specifications, AHJ & Code Compliance, Technical Support, Training & SOP Development

Relevant Coursework: *Thermodynamics, Fluid Dynamics, Materials Science, Electro-Mechanical Systems and Controls, Mechanical Design*

EXPERIENCE

Technical Project Manager

August 2021 - Present

DH Pace Company Inc.

Phoenix, AZ

A national security systems integration company operating across 50+ locations in the United States, delivering engineering solutions across commercial, industrial, and institutional sectors.

Engineering

- Developed circuit layouts and door schematics detailing hardware placement and compatibility.
- Authored an internal software tool to automate telemetry signal verification, eliminating manual review to increase accuracy and reallocate human effort.
- Engineered automated reminder system alerting internal teams of license expirations, enabling proactive customer outreach and contributing to \$1M+ in annual revenue retention across multiple offices.
- Commissioned and integrated systems end-to-end, resolved hardware and software incompatibilities across migrations, and provided technical support to clients across multiple platforms, resolving most issues same-day.
- Validated access control wiring against datasheet schematics prior to field deployment, eliminating field errors and time.

Project Management

- Led 20+ concurrent installation projects from proposal handoff through final city inspection, serving as the single point of contact across customers, onsite technicians, and third-party vendors.

PROJECTS

Self-driving Trashcan Robot | SolidWorks, DFA, 3D Printing, Embedded Systems, C++

<https://khathien.github.io/Trashbot/>

- Designed and built a differential-drive robot platform using repurposed hoverboard BLDC motors to transport trash cans from backyard to curbside on scheduled trash days. Phase 1 (RF controlled) complete; Phase 2 (fully autonomous) in development.
- Implemented ESP-NOW wireless communication between ESP32 controllers, programmed PWM-based speed and direction control for brushless DC motors, and developed C++ firmware with joystick input logic.
- Designed custom 3D printed components using Design for Assembly (DFA) principles, optimizing for component fit, print cost reduction, and ease of repeated disassembly during iterative development.

Canine Dummy for Pet Harness Car Crash Test | SolidWorks, Machinery and Welding, Mechanical Testing

2020-2021

- Led a team of 4 to design and fabricate a metal canine crash test dummy per specifications provided by Exponent, an engineering consultancy; engineered a 30ft drop test in a school arena to simulate crash impact conditions.

ASME Human Powered Vehicle - Recumbent Tricycle | SolidWorks, FEA, DFM, Fabrication

2020-2021

- Led a team of 10 to design, perform FEA analysis, and fabricate Grand Canyon University's first ever recumbent tricycle for ASME competition.

EDUCATION

Grand Canyon University

Phoenix, AZ | April 2021

Bachelor of Science, Mechanical Engineering, GPA: 3.99

Arizona State University

Tempe, AZ | In Progress

Bachelor of Science, Computer Science

Completed courses: *Data Structures, Computational Algorithms, Object-Oriented Programming, Digital Logic*